

Portfolio Project Proposal

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Project 1: Film x Multi-Platform

Judul Akademik + Tech: "Cross-Platform Rating Disparities of Oscar-Nominated Films Using Public Review Data from IMDb, Rotten Tomatoes, and Letterboxd — Python Web Scraping, Sentiment Analysis, and Interactive Visualization"

Sub-Judul: A Quantitative Comparative Study of Audience vs. Critic Preferences Based on Aggregated Multi-Source Review Data Using Pandas, Matplotlib, and Plotly Dash

Deskripsi Naratif: - **Objek:** Semua film yang dinominasikan Oscar 10 tahun terakhir. - **Data:** Review rating, review text, dan user votes dari IMDb, Rotten Tomatoes, Letterboxd. - **Metode:** - Data Cleaning & Processing (Python + Pandas) - Sentiment Analysis (TextBlob / Vader) - Visualisasi Interaktif (Plotly Dash / Tableau) - Analisis Perbedaan Rating Audience vs Critic - **Outcome:** Dashboard interaktif, visualisasi per film, genre, tahun rilis, dengan highlight film rating anomali.

Project 2: Stock Market x Sentiment Analysis

Judul Akademik + Tech: "Sentiment-Driven Price Behavior Analysis of IDX-Listed Stocks Using Social Media Text Mining, Python, and Time-Series Market Data Visualization"

Sub-Judul: An Event-Based Empirical Study on the Relationship Between Retail Investor Sentiment and Short-Term Stock Price Movements Using NLP, Pandas, and Matplotlib

Deskripsi Naratif: - **Objek:** Saham big-cap / blue-chip IDX (BBCA, UNVR, TLKM). - **Data:** - Text: Twitter, Stockbit forum (public) - Price: IDX Historical Price, OHLC, Volume - **Metode:** - NLP & Text Mining (Sentiment Scoring) - Time Series Analysis (Pandas, Matplotlib) - Correlation Analysis: Sentiment vs Price Movement - Interactive Dashboard (Plotly / Power BI) - **Outcome:** Insights terkait pengaruh sentimen retail investor terhadap harga saham harian, event-driven analysis.

Project 3: AI Trend x Tech Stocks

Judul Akademik + Tech: "Impact of Global AI Search Trends on Technology Stock Market Performance Using Google Trends, Python, and Multi-Market Time Series Analysis"

Sub-Judul: A Cross-Correlation and Event Study of AI-Related Keyword Trends and Price Movements in the U.S. Technology Sector Using Scikit-Learn, Statsmodels, and Interactive Visualization

Deskripsi Naratif: - **Objek:** Saham sektor teknologi NASDAQ / S&P 500 (AAPL, MSFT, NVDA). - **Data:** - Google Trends: AI-related keywords - Stock Price: Daily OHLC, Volume via Yahoo Finance API - **Metode:** - Time Series Analysis (Pandas, Statsmodels) - Cross-Correlation & Event Study (AI event vs price movement) - Interactive Dashboard Visualization (Plotly / Power BI) - **Outcome:** Insight pengaruh tren AI global terhadap harga saham teknologi, cocok untuk investor / analis.

Tools & Tech Stack Summary: - Python (Pandas, Matplotlib, Plotly, Scikit-Learn, Statsmodels, TextBlob/Vader) - Data Sources: IMDb, Rotten Tomatoes, Letterboxd, IDX, Twitter, Google Trends, Yahoo Finance API - Dashboard & Visualization: Plotly Dash, Power BI, Tableau - Methodology: Quantitative Comparative, Time-Series Analysis, Sentiment Analysis, Event Study, Cross-Correlation

Portfolio Goal: Menunjukkan kemampuan analisis data, pengolahan multi-source dataset, implementasi teknik statistik & machine learning sederhana, serta presentasi interaktif melalui dashboard. Cocok untuk aplikasi magang / pekerjaan entry-level di IDX, BAE, fintech, atau perusahaan analisis data dan investasi.

Prepared by: Helena Dewi Hapsari **Date:** January 12, 2026